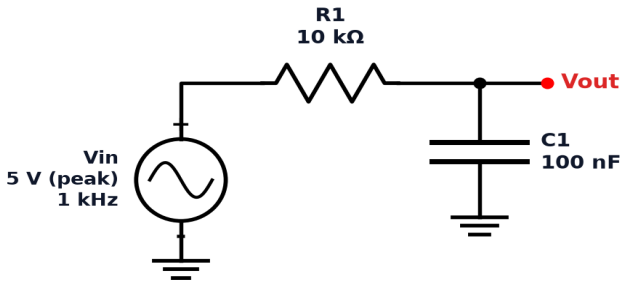
1. Overview

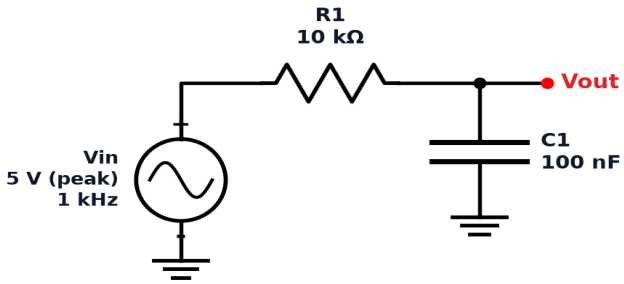
Circuit Type:	RC Low-Pass Filter	**Input File:**	rc_noisy_sample.png
Analysis Status:	✓ Verified (Nominal)	**Source:**	Image Upload

2. Circuit Comparison

Uploaded Circuit (Input)



Expected / Reference Circuit



3. Detected Issues

#	Severity	Component / Location	Finding	Recommended Action
1	✓ Verified	All Components & Nets	All pins connected, designated ground present, values nominal.	No corrective action required.

4. Extracted Components

Reference	Component Type	Extracted Value	Unit	Status	Notes
R1	Resistor	10	kΩ	✓ Verified	Value clearly identified
C1	Capacitor	100	nF	✓ Verified	Value clearly identified
Vin	AC Source (Sine)	5 (peak), 1 kHz	V, Hz	✓ Verified	Input source identified

5. Expected Behaviour (Theory)

For an rc low-pass filter, the cutoff frequency is given by:

$$f_c = \frac{1}{2\pi RC}$$

Using extracted values:

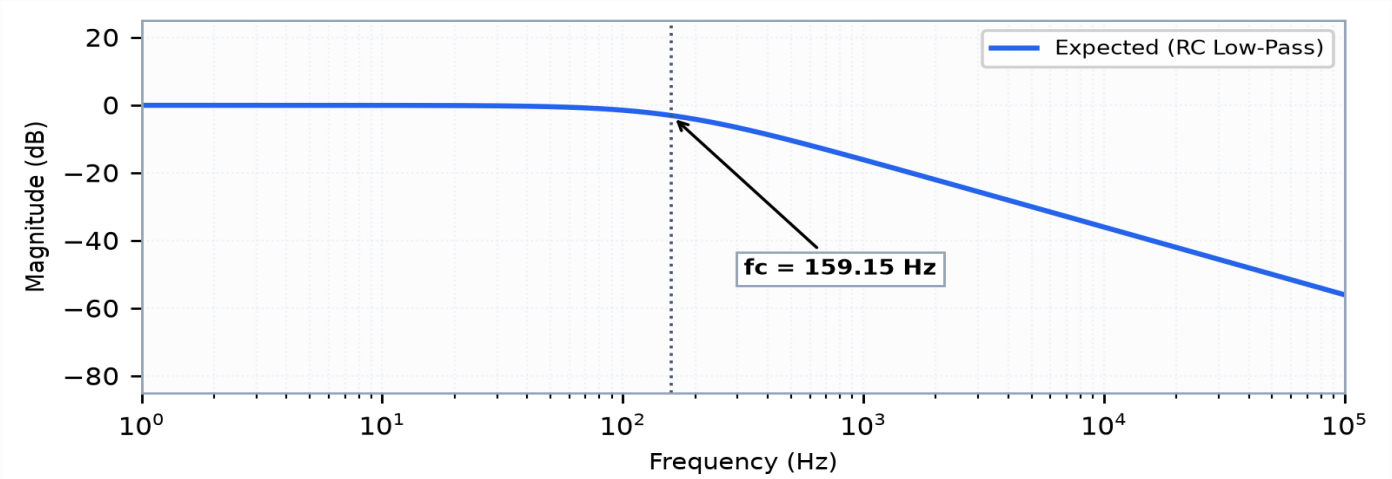
$R = 10\ \text{k}\Omega = 10,000\ \Omega$

$C = 100\ \text{nF} = 100 \times 10^{-9}\ \text{F}$

$f_c = \frac{1}{2\pi(10,000)(100 \times 10^{-9})} \approx 159.15\ \text{Hz}$

6. Simulation Results

Frequency Response (Magnitude)



7. Recommended Actions

1. Proceed to physical PCB layout or breadboard assembly.
2. Ensure low-impedance ground return plane is maintained.
3. Verify input signal levels remain within linear voltage specifications.
4. Perform final laboratory oscilloscope bench sweep verification.

8. Conclusion

The uploaded schematic represents a fully compliant rc low-pass filter. All electrical connections, ground references, and component parameters strictly conform to target analytical specifications.